

View all member:

```
=== MEMBER MANAGEMENT SYSTEM ===
1. Register New Member
2. View All Members
3. Search Member by Email/Phone
4. Update Membership Status (Activate/Deactivate)
5. Back to Main Menu
Select an option: 2

--- All Members ---
ID: 1 | Name: dhanush | Email: dhanush@mail.com | Status: Active
ID: 2 | Name: anil | Email: anil@mail.com | Status: Inactive
ID: 3 | Name: ramu | Email: ramu@mail.com | Status: Active
```

View members in table:

MemberId	Name	Email	Phone	Membership	IsActive	CreatedAt
1	dhanush	dhanush@mail.com	9876543210	1	t	2026-05-18 15:03:16.300884
2	anil	anil@mail.com	9123456780	3	f	2026-05-19 16:56:21.273875
3	ramu	ramu@mail.com	9708654321	2	t	2026-05-26 10:10:18.220958

(3 rows)

Search member by email:

```
=== MEMBER MANAGEMENT SYSTEM ===
1. Register New Member
2. View All Members
3. Search Member by Email/Phone
4. Update Membership Status (Activate/Deactivate)
5. Back to Main Menu
Select an option: 3
Enter Email or Phone number to search: ramu@mail.com

Found Member -> ID: 3, Name: ramu, Email: ramu@mail.com, Phone: 9708654321, Type: Student, Status: Active
```

Search member by phone:

```
=== MEMBER MANAGEMENT SYSTEM ===
1. Register New Member
2. View All Members
3. Search Member by Email/Phone
4. Update Membership Status (Activate/Deactivate)
5. Back to Main Menu
Select an option: 3
Enter Email or Phone number to search: 9876543210

Found Member -> ID: 1, Name: dhanush, Email: dhanush@mail.com, Phone: 9876543210, Type: Basic, Status: Active
```

Book Management:

```
Library management system
```

- 1.Member Management
- 2.Book Management
- 3.Borrow Book Management
- 4.Fine Management
- 5.Reports
- 6.Exit

```
enter your choice:
```

```
2
```

```
=== CATALOG & BOOK MANAGEMENT ===
```

1. Add New Book
2. Add Copies to Existing Book Inventory
3. View Active/Available Catalog Items
4. Search Books Catalog
5. Flag Book Copy as Unavailable
6. Back to Main Menu

Add book:

```
=== CATALOG & BOOK MANAGEMENT ===
```

1. Add New Book
2. Add Copies to Existing Book Inventory
3. View Active/Available Catalog Items
4. Search Books Catalog
5. Flag Book Copy as Unavailable
6. Back to Main Menu

```
Select an option: 1
```

```
Enter Book Title: donet programming
```

```
Enter Author Name: suresh
```

```
Enter Category ID Code: 2
```

```
Master book record appended to catalog successfully!
```

Add Copies to the book:

```
=== CATALOG & BOOK MANAGEMENT ===
1. Add New Book
2. Add Copies to Existing Book Inventory
3. View Active/Available Catalog Items
4. Search Books Catalog
5. Flag Book Copy as Unavailable
6. Back to Main Menu
Select an option: 2
Enter Target Catalog Book ID: 4
Enter Quantity of Physical Copies to Add: 6

Successfully generated 6 inventory tracks for Book ID 4.
```

Books in the database:

```
[library_management=# select * from "Books";
 BookId | Title | Author | CategoryId
-----+-----+-----+-----
      2 | Game of thrones | russo brothers | 1
      3 | oops through cpp | dhanush | 2
      4 | donet programming | suresh | 2
(3 rows)
```

Existing book categories:

```
[library_management=# select * from "BookCategories";
 CategoryId | Name
-----+-----
          1 | Fiction
          2 | Technology
          3 | Science
          4 | Biography
(4 rows)
```

Added bookcopies in the database:

```
[library_management=# select * from "BookCopies";
 BookCopyId | BookId | SerialNumber | Status
```

BookCopyId	BookId	SerialNumber	Status
7	3	SN-3-2-95AA	1
8	3	SN-3-3-A735	1
9	3	SN-3-4-0EC0	1
10	3	SN-3-5-DF17	1
6	3	SN-3-1-5D53	4
11	4	SN-4-1-6299	1
12	4	SN-4-2-9D19	1
13	4	SN-4-3-1EE4	1
14	4	SN-4-4-C379	1
15	4	SN-4-5-C934	1
16	4	SN-4-6-0D51	1

Available Copies

```
--- Displaying Available Catalog Material ---
ID: 3 | Title: oops through cpp | Author: dhanush [Copies Available: 4]
ID: 4 | Title: donet programming | Author: suresh [Copies Available: 6]
```

Search books based on author:

```
Select an option: 4
Enter your lookup parameter (Title or Author match context): suresh

--- Search Matches matching criteria: 'suresh' ---
ID: 4 | Title: donet programming | Author: suresh [Copies Available: 6]

=== CATALOG & BOOK MANAGEMENT ===
1. Add New Book
2. Add Copies to Existing Book Inventory
3. View Active/Available Catalog Items
4. Search Books Catalog
5. Flag Book Copy as Unavailable
6. Back to Main Menu
Select an option: 4
Enter your lookup parameter (Title or Author match context): donet programming

--- Search Matches matching criteria: 'donet programming' ---
ID: 4 | Title: donet programming | Author: suresh [Copies Available: 6]
```

Book copies in Database:

```
[library_management=# select * from "BookCopies";
```

BookCopyId	BookId	SerialNumber	Status
7	3	SN-3-2-95AA	1
8	3	SN-3-3-A735	1
9	3	SN-3-4-0EC0	1
10	3	SN-3-5-DF17	1
6	3	SN-3-1-5D53	4
12	4	SN-4-2-9D19	1
13	4	SN-4-3-1EE4	1
14	4	SN-4-4-C379	1
15	4	SN-4-5-C934	1
16	4	SN-4-6-0D51	1
11	4	SN-4-1-6299	4

(11 rows)

Fine payment History of a member:

1. View Pending Fines of a Member
2. Pay Fine
3. View Fine History Ledger
4. Return to Main Desk Menu

=====

enter the choice 3

--- View Fine History ---

Enter Cardholder Member ID: 1

=====

Log Item ID	Posting Date	Amount Charge	Status
#3	27/05/2026 7:31:45 AM	₹60.00	CLEARED / PAID

=====

Press any key to continue...

Cannot issue a book when the member account is suspended:

--- Issue New Book Loan ---

Enter Cardholder Member ID: 1

Enter Master Book ID to find copy: 3

Fatal Processing Defect: Transaction Denied: This membership account is currently suspended or inactive.

View Borrowed books:

--- Report Output: Option 1 ---

Copy ID	Book Title	Borrowed By	Due Date
#7	oops through cpp	dhanush	03/06/2026

Press any key to return to reports menu...

Book Management Screen:

```
=====
1. Books Currently Borrowed
2. Overdue Books Tracker
3. Members with Pending Fines Summary
4. Top 5 Most Borrowed Titles
5. Available Book Stock Counts by Category
6. Individual Member Lending History Audit
7. Return to Main Control Panel
=====
```

Enter the choice

Top 5 books borrowed :

--- Report Output: Option 4 ---

Book ID	Book Title	Times Loaned
#3	oops through cpp	2

checkouts

Press any key to return to reports menu...

Available books count:

--- Report Output: Option 5 ---

Genre / Category	In-Stock Count
Technology	9

copies

Press any key to return to reports menu...

Individual member book borrowing history:

```
--- Report Output: Option 6 ---
Enter Member ID to query: 1

Lending Audit Trail Ledger For: dhanush (ID: #1)
=====
Copy ID | Book Title | Checkout Date | Return Date | Status
=====
#7 | oops through cpp | 27/05/2026 | ACTIVE LOAN | Active
#6 | oops through cpp | 19/05/2026 | 27/05/2026 | Returned
Press any key to return to reports menu...
```

View Customers table and orders table:

```
sql_practice=# select * from customers;
 id | name | email | city | created_at
-----+-----+-----+-----+-----
  1 | Alice Smith | alice@example.com | New York | 2023-01-15
  2 | Bob Johnson | bob@example.com | Chicago | 2023-02-20
  3 | Charlie Brown | charlie@example.com | Houston | 2023-03-10
  4 | Diana Prince | diana@example.com | Seattle | 2023-04-05
(4 rows)
```

```
sql_practice=# select * from orders;
 id | customer_id | total | status | ordered_at
-----+-----+-----+-----+-----
  1 | 4 | 150.00 | completed | 2024-01-15
  2 | 2 | 200.50 | completed | 2024-01-22
  3 | 2 | 300.00 | pending | 2024-01-25
  4 | 3 | 450.00 | completed | 2024-02-14
  5 | 1 | 100.00 | completed | 2023-12-28
  6 | 3 | 600.00 | completed | 2024-03-10
  7 | 3 | 120.00 | cancelled | 2024-03-15
  8 | 3 | 250.00 | completed | 2024-03-20
(8 rows)
```